NAVRACHANA UNIVERSITY

Teaching Plan

Software Engineering (CS330)

Credit : 3

<table>
<tr><td>Faculty</td><td>
<table>
<tr><th>Name</th><th>Contact</th><th>Office Hours</th><th>Office Location</th></tr>
<tr><td>Ujwala Petigara</td><td>ujwala.petigara@nuv.ac.in</td><td></td><td></td></tr>
</table>
</td></tr>
<tr><td>School</td><td>SET</td></tr>
<tr><td>Semester</td><td>Autumn (July to Dec 2025)</td></tr>
<tr><td>Lecture time & Weedays</td><td>
<table>
<tr><td>SECTION_1</td><td>Wednesday</td><td>14:00:00-15:00:00</td><td>07-07-2025 TO 14-11-2025</td></tr>
<tr><td></td><td>Thursday</td><td>12:00:00-13:00:00</td><td>07-07-2025 TO 14-11-2025</td></tr>
<tr><td></td><td>Monday</td><td>12:00:00-13:00:00
12:00:00-13:00:00</td><td>07-07-2025 TO 07-09-2025
08-09-2025 TO 14-11-2025</td></tr>
<tr><td>SECTION_2</td><td>Wednesday</td><td>10:00:00-11:00:00</td><td>07-07-2025 TO 14-11-2025</td></tr>
<tr><td></td><td>Friday</td><td>12:00:00-13:00:00</td><td>07-07-2025 TO 14-11-2025</td></tr>
<tr><td></td><td>Tuesday</td><td>11:00:00-12:00:00
11:00:00-12:00:00</td><td>07-07-2025 TO 07-09-2025
08-09-2025 TO 14-11-2025</td></tr>
</table>
</td></tr>
<tr><td>Pre-requisite</td><td></td></tr>
<tr><td>Co-requisite</td><td></td></tr>
<tr><td>Anti-requisite</td><td></td></tr>
<tr><td>Course Description</td><td>This course is aimed at helping students build up an understanding of how to develop a software system from scratch by guiding them thru the development process and giving them the fundamental principles of system development with object oriented technology using UML. The course will initiate students to the different software process models, project management, software requirements engineering process, systems analysis and design as a problem-solving activity, key elements of analysis and design, and the place of the analysis and design phases within the system development life cycle</td></tr>
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<tr><td>Course Objectives</td><td>This course is aimed at helping students build up an understanding of how to develop a software system from scratch by guiding them thru the development process and giving them the fundamental principles of system development with object oriented technology using UML. The course will initiate students to the different software process models, project management, software requirements engineering process, systems analysis and design as a problem-solving activity, key elements of analysis and design, and the place of the analysis and design phases within the system development life cycle</td></tr>
<tr><td>Course Outcomes</td><td>Student will be able to
1. To write and survey solution for multidisciplinary case-study to choose appropriate Process model.
2. To design various system models with respect to different approaches.
3. To implicate various design specification of software.
4. To perform Cost/Effort estimation Planning.
5. To write different test cases according to system design.</td></tr>
<tr><td>Course Typology</td><td>Theory</td></tr>
<tr><td>Course Outline</td><td>As mentioned below in detailed Session Plan below</td></tr>
<tr><td>Pedagogy</td><td>Theoretical Foundations, Software Development Practices, Collaborative Learning, Continuous Assessment and Feedback</td></tr>
<tr><td>Expectations from Students</td><td>Students should showed the interest in the subject and try to understand the concepts thro’ regular revision and attending labs and lectures 100 % and practice programs given in assignments</td></tr>
<tr><td>Assessment / Evaluation</td><td></td></tr>
<tr><td>Project / Assignment Details</td><td>Students should showed the interest in the subject and try to understand the concepts thro’ regular revision and attending labs and lectures 100 % and practice programs given in assignments</td></tr>
<tr><td>Course Material</td><td>Text books:
1. Roger Pressman, “Software Engineering: A Practitioners Approach”, 6thEdition, McGraw Hill, 2010.
2. Ian Somerville, “Software Engineering”,9 th Edition, Addison Wesley, 2011
3. James F. Peters and WitoldPedrycz, “Software Engineering – An Engineering Approach”, Wiley, 2000.
Reference books:
1. Roger S.Pressman, Software engineering- A practitioner’s Approach, McGraw-Hill International Editions
2. Ian Sommerville, Software engineering, Pearson education Asia
3. Pankaj Jalote, Software Engineering – A Precise Approach Wiley
4. Software Engineering Fundamentals by Ali Behhforoz & Frederick Hudson OXFOR</td></tr>
<tr><td>Additional Information</td><td>The course will be useful to understand and implement work on project</td></tr>
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Session Plan	Topic Title	Session No.	Topic & Subtopic Details	Readings, Cases, etc.	Activities	Important Dates
		0				
	INTRODUCTION	1	Introduction to Software Engineering Software Components, 8 Software Characteristics, Software Engineering Processes, Similarity and Differences from Conventional Engineering Processes			
		2	Software Quality Attributes. Software Development Life Cycle (SDLC) ModelsWater Fall Model, Prototype Model Spiral Model			
		3	Evolutionary Development Models,Iterative Enhancement Models			
		4	Typical Application for each model, Agile methodology, Process and Project Metrics			
	SYSTEM ANALYSIS	5	System Requirement			
		6	System Requirement			
		7	Stakeholders			
		8	Techniques for information gathering			
		9	Techniques for information gathering			
		10	Validating the requirements			
		11	Validating the requirements			
		12	Feasibility Analysis			
		13	Tests for feasibility			
		14	Cost Benefit Analysis			
		15	Feasibility report			
		16	The system Proposal Data flow diagram and UML analysis diagrams			
		17	The system Proposal Data flow diagram and UML analysis diagrams			
	SOFTWARE DESIGN	18	Software Design Specification– Abstraction Modularity			
		19	Effective modular design			
		20	Cohesion and Coupling			
		21	Example of code for cohesion and coupling			
		22	User Interface Design – Human Factors,			
		23	Interface standards– User Interface Design Process Application Architecture for Information System			
		24	Interface standards– User Interface Design Process Application Architecture for Information System			
		25	Deployment using UML diagrams			
		26	Component and deployment diagram for various architectures Design Patterns Gang of four patterns			
	SOFTWARE QUALITY ASSURANCE	27	Software estimation			
		28	Empirical estimation models			
		29	Cost/Effort estimation Planning			
		30	Work breakdown Structure, Gantt Chart			
		31	Evaluate schedule and cost variance Risk Management Risk Identification			
		32	Risk Assessment			
		33	Risk Projection			
		34	RMMM			
		35	Software Configuration Management			
		36	SCM process, version and change control Quality metrics			
	SOFTWARE TESTING	37	Software Testing Basic concept and terminology			
		38	Verification & validation			
		39	White Box Testing-, Black Box Testing			
		40	Integration, Validation and system testing			
		41	Object Oriented testing methodsClass Testing, Interclass testing			
		42	Object Oriented testing methodsClass Testing, Interclass testing			
		43	Interclass testing,			
		44	testing architecture,			
		45	Behavioural testing			
Publish Status	Published					

NAVRACHANA UNIVERSITY

Teaching Plan

Core Java Programming (CS348)

Credit : 2

Faculty	<table><tr><td>Name</td><td>Contact</td><td>Office Hours</td><td>Office Location</td></tr><tr><td>Vivek Bhatt</td><td>vivek.bhatt@nuv.ac.in</td><td></td><td></td></tr></table>
School	SET
Semester	Spring (Jan to June 2025)
Lecture time & Weedays	<table><tr><td>SECTION_1</td><td>Monday</td><td>10:00:00-11:00:00</td><td>08-01-2025 TO 09-05-2025</td></tr><tr><td></td><td>Tuesday</td><td>10:00:00-11:00:00</td><td>08-01-2025 TO 09-05-2025</td></tr><tr><td>SECTION_2</td><td>Tuesday</td><td>12:00:00-13:00:00</td><td>08-01-2025 TO 09-05-2025</td></tr><tr><td></td><td>Wednesday</td><td>12:00:00-13:00:00</td><td>08-01-2025 TO 09-05-2025</td></tr></table>
Pre-requisite	
Co-requisite	
Anti-requisite	
Course Description	This course of study builds on the skills gained by students in Java Fundamentals or Java Foundations to help advance Java programming skills. Students will design object-oriented applications with Java and will create Java programs using hands-on, engaging activities.
Course Abstract	Through a combination of theoretical understanding and practical exercises, the course equips students with the skills to design efficient, scalable, and dynamic Java-based applications, preparing them for both academic and professional challenges in software development.
Course Objectives	1. Introduce Java as a programming language. 2. Introduce Java as an Object-Oriented programming language. 3. Introducing the Concepts of Exception Handling and Multithreading 4. Introducing the Concepts of Java Collections Framework and Database Programming 5. Introducing the concepts of Swing GUI and Servlets.
Course Outcomes	CO1. Learn the Java programming language: its syntax, idioms, patterns, and styles CO2.The skills to apply OOP concepts in Java programming in problem solving. CO3. Understand the concepts of Exception Handling and Multithreading CO4. Learn the essentials of the Java Collections Framework and Database Programming CO5. Learn Swing GUI and Servlets
Course Typology	Theory
Course Outline	Unit - 1 : Java Basics – History of Java, Java Features, comments, data types, variables, constants, scope and life time of variables, operators, operator hierarchy, expressions, type conversion and casting, enumerated types, control flow-block scope, conditional statements, loops, break and continue statements, simple java program, arrays, input and output, formatting output, Review of OOP concepts, constructors, methods, static fields and methods, access control, this reference, overloading methods and constructors, recursion, garbage collection, building strings, exploring string class, Enumerations, Generics Unit - 2 : Object Oriented Concepts – Inheritance concept, benefits of inheritance, Super classes and Sub classes, Member access rules, Inheritance hierarchies, super uses, preventing inheritance: final classes and methods, casting, polymorphism-dynamic binding, method overriding, abstract classes and methods, the Object class and its methods. Unit - 3 : Exception handling and Multithreading– Dealing with errors, benefits of exception handling, the classification of exceptions-exception hierarchy, checked exceptions and unchecked exceptions, usage of try, catch, throw, throws and finally, re-throwing exceptions, exception specification, built in exceptions, creating own exception sub classes, Multithreading – Differences between multiple processes and multiple threads, thread states, creating threads, interrupting threads, thread priorities, synchronizing threads, interthread communication, thread groups, daemon thread Unit - 4 : Java Collections Framework and Database Programming – Introduction to Java Collections, Overview of Java Collection frame work, Commonly used Collection classes – Array List, Linked List, HashSet, HashMap, TreeMap, Collection Interfaces –, Using javautil Files – streams- byte streams, character streams, text Input/output, binary input/output, random access file operations, File management using File class, Using javaio., Database Programming using RDBMS (MySQL), Developing CRUD applications with CLI, GUI and Web Applications. Unit - 5 : Swing GUI, Servlets and JSP Package Organizing Classes and Interfaces in Packages , Package as Access Protection , Defining Package ,CLASSPATH Setting for Packages , Making JAR Files for Library Packages Import and Static Import Naming Convention For Packages. Event-Driven Programming in Java, Event- Handling Process, Event Handling Mechanism, The Delegation Model of Event Handling, Event Classes, Event Sources, Event Listeners, Adapter Classes as Helper Classes in Event Handling. Desktop Applications with Swing/AWT. Web Application Development using Java, Java Servlet, Java Server Pages, MVC Development, Introduction to Spring Framework.
Pedagogy	Flipped classrooms, Presentation, Real-time problem solving with Java programms
Expectations from Students	Laptops with JDK (at least 1.8) installed
Assessment / Evaluation	
Project / Assignment Details	Continuous Evaluation (60%) 1. Assignment (OOPS concepts + Multi-threading) (20%) 2. Assignment (DBMS) (20%) 3. Assignment (Web Applications) (20%) Final Evaluation (40%) Combined project with DBMS course
Course Material	1. The complete reference, 7th edition, Herbert Schildt, TMH. 2. Java for Programmers, PJDeitel and HMDeitel, Pearson education / Java: How to Program PJDeitel and HMDeitel,8th edition, PHI 3. Core Java, Volume 1- Fundamentals, eighth edition, Cay Horstmann and Gary Cornell, Pearson education.
Additional Information	This course requires serious attention to learn a programming language from scratch

Session Plan

Topic Title	Session No.	Topic & Subtopic Details	Readings, Cases, etc.	Activities	Important Dates
Java Basics	1	History of Java, Java Features, comments, data types, variables, constants, scope and life time of variables, operators, operator hierarchy, expressions			
Java Basics	2	Type conversion and casting, enumerated types, control flow-block scope			
Java Basics	3	Conditional statements, loops, break and continue statements, simple java program, arrays, input and output, formatting output			
Java Basics	4	Review of OOP concepts			
Java Basics	5	Constructors, methods, static fields and methods, access control, this reference, overloading methods and constructors, recursion			
Java Basics	6	Garbage collection, building strings, exploring string class, Enumerations, Generics			
Object Oriented Concepts	7	Inheritance concept, benefits of inheritance			
Object Oriented Concepts	8	Super classes and Sub classes, Member access rules, Inheritance hierarchies			
Object Oriented Concepts	9	Super uses, preventing inheritance: final classes and methods			
Object Oriented Concepts	10	Casting, polymorphism-dynamic binding			
Object Oriented Concepts	11	Method overriding, abstract classes and methods			
Object Oriented Concepts	12	The Object class and its methods.			
Exception Handling and Multithreading	13	Dealing with errors, benefits of exception handling			
Exception Handling and Multithreading	14	The classification of exceptions, Exception hierarchy			
Exception Handling and Multithreading	15	Checked exceptions and unchecked exceptions			
Exception Handling and Multithreading	16	Usage of try, catch, throw, throws and finally			
Exception Handling and Multithreading	17	Rethrowing exceptions, exception specification, built in exceptions, creating own exception sub classes			
Exception Handling and Multithreading	18	Multithreading – Differences between multiple processes and multiple threads, thread stat			
Exception Handling and Multithreading	19	Creating threads, interrupting threads, thread priorities, synchronizing threads, interthread communication, thread groups, daemon thread			
Java Collections Framework and Database Programming	20	Introduction to Java Collections, Overview of Java Collection frame work, Commonly used Collection classes – Array List, Linked List, HashSet, HashMap, TreeMap, Collection Interfac			
Java Collections Framework and Database Programming	21	Using javautil Files – streams- byte streams, character streams, text Input/output, binary input/output, random access file operation			
Java Collections Framework and Database Programming	22	File management using File class, Using javaio			
Java Collections Framework and Database Programming	23	Database Programming using RDBMS (MySQL), Developing CRUD applications with CLI, GUI and Web Applications			
Swing GUI, Servlets and JSP	24	Package Organizing Classes and Interfaces in Packages , Package as Access Protection			
Swing GUI, Servlets and JSP	25	Defining Package ,CLASSPATH Setting for Packages			
Swing GUI, Servlets and JSP	26	Making JAR Files for Library Packages Import and Static Import Naming Convention For Packages			
Swing GUI, Servlets and JSP	27	Event-Driven Programming in Java, Event- Handling Process, Event Handling Mechanism			
Swing GUI, Servlets and JSP	28	The Delegation Model of Event Handling, Event Classes, Event Sources, Event Listeners			
Swing GUI, Servlets and JSP	29	Adapter Classes as Helper Classes in Event Handling. Desktop Applications with Swing/AWT			
Swing GUI, Servlets and JSP	30	Web Application Development using Java, Java Servlet, Java Server Pages, MVC Development, Introduction to Spring Framework			

Publish Status	Published

NAVRACHANA UNIVERSITY

Teaching Plan

Database Management Systems (CS237)

Credit : 3

Faculty

Name	Contact	Office Hours	Office Location
Avani Vasant	avani.vasant@nuv.ac.in		

School

SET

Semester

Spring (Jan to June 2025)

Lecture time & Weedays

SECTION_1	Friday	16:00:00-17:00:00 12:00:00-13:00:00	08-01-2025 TO 12-01-2025 13-01-2025 TO 09-05-2025
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Pre-requisite

Co-requisite

Anti-requisite

Course Description

This course introduces students to basic database design and the use of databases in applications. Throughout the course students are introduced to Database and Database Management System, advantages and disadvantages of the different database models, relational model and Structured Query Language (SQL), Understanding constraints, transaction management, ACID properties. It includes the design using various diagrams like ER, Sequence, and Activity. The designed system will produce the 'reports' for specific queries.

Course Abstract

The course covers the fundamental concepts of database design and implementation, relational concepts, data modeling, conceptual and logical database design, use of SQL as a data- definition, manipulation, control language, and database administration.

Course Objectives

• To introduce fundamental concepts of DBMS.
• To design and create the Database model.
• To understand the Normalization techniques to reduce the data Redundancy.
• To Learn the query processing system, recovery techniques and management of transactions in database.
• To empower and expose the students with database implementation using SQL and PL/SQL languages.

Course Outcomes

CO1 To be able to apply analytical skills to map out the conceptual design for a real-life problem and explain the fundamental elements of Database management System.
CO2 To be able to draw the model of Relational Database using Entity Relationship (ER) model
CO3 To be able to Improve the database design by normalization.
CO4 To be able to evaluate the logical design to translate into a specific data model and physical design to meet system storage requirements.
CO5 To be able to evaluate Oracle features and Oracle related products for maintaining the integrity and performance of databases.

Course Typology

Theory

Course Outline

Data and Information: . Introduction to Structured, Semi-structured and unstructured data. Limitations of Manual Data Processing, Advantages of DBMS over traditional computer file-based processing approach, Functions of DBA.
ENTITY-RELATIONSHIP MODEL: Basic concepts, entity, attributes, relationship, Design process, constraints, Keys, weak entity sets, extended E-R features – generalization, specialization, aggregation, Reduction of ER to Relational database.
RELATIONAL MODEL
Structure of relational databases, Domains, Keys - Primary, Secondary and Composite, Types of Constraint, Anomaly in database design, Functional Dependency- definition, type of FDs, closure of FD set, closure of attributes, Equivalence of FD, irreducible set of FD, Decomposition- lossless and Dependency Preservation, Normalization – 1Nf, 2NF, 3NF, BCNF, 4NF and 5NF.
RELATIONAL ALGEBRA AND SQL
Introduction to query language, Fundamental operators and syntax, relational algebra queries. SQL Concepts: Basics of SQL, DDL, DML, DQL, DCL, TCL.
QUERY AND STORAGE STRATEGIES
Query Processing & Query Optimization: Overview, measures of query cost. Storage strategies: Indices, B-trees, hashing.
TRANSACTION MANAGEMENT
Transaction Management: Transaction concepts, properties of transactions, Transaction states, concurrent executions of transactions and related problems, serializability of transactions, testing for serializability, Types of schedules, concurrency control, two-phase locking protocol.
Reference books:
1. Silberschatz, A., Korth, H. F., & Sudarshan, S. (2010). Database system concepts (int'l ed) (6th ed.). McGraw Hill Higher Education.
2. Date, C. J. (1977). An introduction to database systems. Pearson Education.
3. Date, C. J. (2012). Database design and relational theory: Normals forms and all that jazz. O'Reilly Media.
4. Gruber, M. (1990). Understanding SQL.
5. Ivan Bayross. SQL, PL/SQL – The Programming Language of Oracle

Pedagogy

. Teaching Theory and Practice with Active Learning Approach
• Brainstorming
• Stop-and-Think
• Think-Pair-Share
• Exercise Breaks
• Case Study/Group Problem Solving

Expectations from Students

1. Attendance is mandatory as per university norms.
2. Be Regular in attending sessions and reaching sessions on time.
3. To be attentive and participate in classroom discussions.
4. To refrain from using mobile phones in the classroom.
5. Read concepts before the class and revise after the class.
6. Do assigned homework every day; this is the only way to learn and succeed.
7. Spend 5 hours a week reading the course from reference books.
8. Ask for help when appropriate.
9. Maintain academic integrity in assignments.

Assessment / Evaluation

Project / Assignment Details

Will be discussed in classroom and assigned through Teams

Course Material

PPTs, Lecture Notes will be posted progressively.

Additional Information

Students are encouraged
To learn from NPTEL Archive courses on Database Management System

Session Plan

Topic Title	Session No.	Topic & Subtopic Details	Readings, Cases, etc.	Activities	Important Dates
DATA AND INFORMATION	1	Database-System Applications, Purpose of Database Systems, View of Data	Reference Books	Brainstorming	
DATA AND INFORMATION	2	Limitations of Manual Data Processing, Advantages of DBMS over traditional computer file-based processing approach	Reference Books	Discussion	
DATA AND INFORMATION	3	Database Languages, Relational Databases, Database Users and Administrators, Functions of DBA	Reference Books	Learning	
ENTITY-RELATIONSHIP MODEL	4	Basic concepts, entity, attributes, relationship	Reference Books	Think-Pair-Share	
ENTITY-RELATIONSHIP MODEL	5	Design process, constraints, Keys	Reference Books	Learning	
ENTITY-RELATIONSHIP MODEL	6		Reference Books	Brainstorming	
ENTITY-RELATIONSHIP MODEL	7	Reduction of ER to Relational database	Reference Books	Problem Solving	
RELATIONAL MODEL	8	Structure of relational databases, Domains	Reference Books	Active Learning	
RELATIONAL MODEL	9	Keys - Primary, Secondary and Composite	Reference Books	Think-Pair-Share	
RELATIONAL MODEL	10	Types of Constraint	Reference Books	Active Learning	
RELATIONAL MODEL	11	Anomaly in database design, Functional Dependency-definition	Reference Books	Discussion	
RELATIONAL MODEL	12	type of FDs, Decomposition- lossless and Dependency Preservation	Reference Books	Problem Solving	
RELATIONAL MODEL	13		Reference Books	Group Problem Solving	
RELATIONAL MODEL	14		Reference Books	Group Problem Solving	
RELATIONAL MODEL	15		Reference Books	Group Problem Solving	
RELATIONAL MODEL	16		Reference Books	Group Problem Solving	
RELATIONAL MODEL	17		Reference Books	Group Problem Solving	
RELATIONAL MODEL	18		Reference Books	Group Problem Solving	
RELATIONAL MODEL	19	closure of FD set	Reference Books	exercise breaks	
RELATIONAL MODEL	20	closure of attributes	Reference Books	exercise breaks	
RELATIONAL MODEL	21	Equivalence of FD, irreducible set of FD	Reference Books	exercise breaks	
RELATIONAL ALGEBRA AND SQL	22	Introduction to query language, Fundamental operators and syntax	Reference Books	Discussion	
RELATIONAL ALGEBRA AND SQL	23	Introduction to query language, Fundamental operators and syntax	Reference Books	Discussion	
RELATIONAL ALGEBRA AND SQL	24	relational algebra queries	Reference Books	Problem Solving	
RELATIONAL ALGEBRA AND SQL	25	relational algebra queries	Reference Books	Group Problem Solving	
RELATIONAL ALGEBRA AND SQL	26	SQL Concepts: Basics of SQL, DDL, DML, DQL, DCL, TCL	Reference Books	Exercises	
RELATIONAL ALGEBRA AND SQL	27	SQL Concepts: Basics of SQL, DDL, DML, DQL, DCL, TCL	Reference Books	Exercises	
QUERY AND STORAGE STRATEGIES	28	Query Processing & Query Optimization: Overview, measures of query cost	Reference Books	Active Learning / Presentations	
QUERY AND STORAGE STRATEGIES	29	Query Processing & Query Optimization: Overview, measures of query cost	Reference Books	Active Learning / Presentations	
QUERY AND STORAGE STRATEGIES	30	Query Processing & Query Optimization: Overview, measures of query cost	Reference Books	Active Learning / Presentations	
QUERY AND STORAGE STRATEGIES	31	Storage strategies: Indices, B-trees, hashing	Reference Books	Active Learning / Presentations	
QUERY AND STORAGE STRATEGIES	32	Storage strategies: Indices, B-trees, hashing	Reference Books	Active Learning / Presentations	
QUERY AND STORAGE STRATEGIES	33	Storage strategies: Indices, B-trees, hashing	Reference Books	Active Learning / Presentations	
QUERY AND STORAGE STRATEGIES	34	Storage strategies: Indices, B-trees, hashing	Reference Books	Active Learning / Presentations	
TRANSACTION MANAGEMENT	35	Transaction Management: Transaction concepts, properties of transactions	Reference Books	Brainstorming & Discussion	
TRANSACTION MANAGEMENT	36	Transaction Management: Transaction concepts, properties of transactions	Reference Books	Brainstorming & Discussion	
TRANSACTION MANAGEMENT	37	Transcation states, Schedules	Reference Books	Exercises	
TRANSACTION MANAGEMENT	38	concurrent executions of transactions and related problems	Reference Books	Case study/ Group Problem Solving	
TRANSACTION MANAGEMENT	39	concurrent executions of transactions and related problems	Reference Books	Case study/ Group Problem Solving	
TRANSACTION MANAGEMENT	40	serializability of transactions	Reference Books	Case study/ Group Problem Solving	
TRANSACTION MANAGEMENT	41	testing for serializability	Reference Books	Brainstorming & Discussion	
TRANSACTION MANAGEMENT	42	Types of schedule	Reference Books	Exercises	
TRANSACTION MANAGEMENT	43	concurrency control	Reference Books	Exercises	
TRANSACTION MANAGEMENT	44	two-phase locking protocol	Reference Books	Think-Pair-Share	
Future Directions & Reflections	45	Emerging Trends, Research, or Advancements in the field of Database Management, Personal Insights and Learning Outcome		Discussions/ Sharing/ Group Sharing	

Publish Status	Published